Feature-Level Data Fusion Based Deep Learning for Soil Moisture Prediction Using Multi-Source Satellite Data

Pyingkodi M
Kongu Engineering College
Erode, India
pyingkodi@kongu.ac.in

Danushree V S
Kongu Engineering College
Erode, India
danushreevs.25mca@kongu.edu

Duraimurugan K
Kongu Engineering College
Erode, India
duraimurugank.25mca@kongu.edu

Krupa N S
Kongu Engineering College
Erode, India
krupans.25mca@kongu.edu

***Abstract* — Soil moisture plays a crucial role in the areas of agriculture, hydrological management, and observation of the environment. The assessment of soil moisture helps agricultural experts develop more accurate ways of improving the productivity of crops through enhanced irrigation systems. The conventional approaches for the assessment of soil moisture have been based on the deployment of terrestrial sensors and manual measurement techniques, which have been associated with substantial economic costs and limited spatial extension capabilities. This manuscript presents the implementation of the feature-level data fusion-based deep learning approach for the estimation of soil moisture using multi-source satellite imagery data sets. Environmental and satellite image-derived characteristics such as NDVI, NDMI, precipitation, land surface temperature (LST), topographical elevations, topographical slope, and radar backscatter (VV and VH) have been integrated for the development of accurate predictions of soil moisture using various machine learning approaches such as Random Forest and XGBoost algorithms and the Deep Neural Network (DNN).In order to enhance the overall performance, the ensemble method is used for the integration of the predictions made by the models. There are four categories of the soil moisture levels, namely, Dry, Low, Medium, and High. According to the moisture content of the soil, the crop selection module suggests the crops that can be grown. In order to enhance the agricultural decisions, the system produces the soil moisture maps for the region of Tamil Nadu. Experimental results show that the proposed ensemble model achieves a Root Mean Square Error (RMSE) of 0.00526, a Mean Absolute Error (MAE) of 0.00339, and an R^2 score of 0.9904, demonstrating the effectiveness and reliability of the approach for accurate large-scale soil moisture prediction.**

Keywords*— *Soil Moisture Prediction, Feature-Level Data Fusion, Deep Learning, Multi-Source Satellite Data, Precision Agriculture.

I. INTRODUCTION

Soil moisture is an important aspect of various activities related to agriculture and water resources. Soil moisture is also used for environmental applications. Soil moisture has a direct impact on crops, irrigation, and drought. Estimation of soil moisture is extremely important to assist farmers in taking proper decisions regarding the selection of crops. This would help to improve the growth of crops. Estimation of soil moisture is generally done by using traditional measurement techniques. In the traditional measurement technique, it is necessary to install sensors in the soil. Estimation of soil moisture by using the traditional measurement technique is time-consuming and costly. It is not possible to monitor the soil moisture of a large geographic area like Tamil Nadu by using the traditional measurement technique. In recent times, remote sensing techniques have been used to monitor various environmental activities. Data like Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), rainfall, land surface temperature (LST), elevation, slope, and radar backscatter values obtained by satellites are extremely useful for the purpose of monitoring the soil moisture. But it is a big challenge to utilize the data.

The deep learning and machine learning approaches have been proven to have a high potential for analyzing large-scale environmental data, where accurate predictions can be made by identifying important patterns from the data. The Random Forest, XGBoost, and Deep Neural Network (DNN) are capable of efficiently learning the complex relationships between multiple features, enabling accurate predictions to be made. However, the use of these models individually might not efficiently exploit the information from the multiple source satellite data, leading to less reliable predictions.

In light of the aforementioned problems, the study proposes a feature-level data fusion-based deep learning model for soil moisture prediction by utilizing the information from multiple source satellite and environmental data. In the proposed model, multiple satellite-based features such as NDVI, NDMI, rainfall, land surface temperature, elevation, slope, and radar backscatter values such as VV and VH are combined at the feature level, after which machine learning models such as Random Forest, XGBoost, and a Deep Neural Network are employed to predict the soil moisture levels accurately.

In order to improve the performance of the prediction, the suggested approach utilizes a technique that combines the

predictions of various models. Additionally, various groups of expected soil levels are used in the system, which include Dry, Low, Medium, and High. This may help the farmer understand the state of the soil. The farmer will be able to receive recommendations from the crop recommendation module depending on the state of the soil.

II. LITERATURE REVIEW

Chen et al. [1] proposed an advanced multisource satellite data fusion system to trace the alfalfa plants with the help of Landsat 7, Landsat 8, Sentinel-2, and Sentinel-1 satellite data. Three types of fusion models were used: Landsat2Sentinel-2, SAR2Optical, and Optical2SAR. In order to improve the fusion of the spectral bands and vegetative indexes, the authors used the Random Forest model. In this regard, the proposed method significantly reduced the RMSE compared to the traditional linear regression models. In the paper, it is clearly demonstrated that the precision of the crop monitoring is improved with the help of satellite data.

Pandey et al. [2] have developed an operational framework for predicting high-resolution surface soil moisture for Indian croplands using EOS-04 satellite data. This study resulted in a 500 m resolution soil moisture product using Synthetic Aperture Radar and an active-passive technique. In-situ measurements were used to validate the suggested model for various agro-climatic regions in India. The results showed good agreement with in-situ measurements, with ubRMSE ranging from 0.051 to 0.078 m^3/m^3. This architecture highlights the use of SAR for accurate soil moisture monitoring in various applications.

Dinesh et al. [3] developed a machine learning technique to estimate soil moisture from simulated NASA-ISRO SAR (NISAR) L-band SAR data. The research utilized various decomposition techniques to obtain polarimetric information. A variety of machine learning techniques were used in this research, with Random Forest being the best performer. When considering vegetation effects, this model performed well in soybean crop estimation with R^2 up to 0.92 and an RMSE of 0.042. The Water Cloud Model was also used in this research to improve soil moisture prediction and dielectric constants. The research highlights the potential of machine learning techniques using SAR images for upcoming soil moisture monitoring applications.

Usmani et al. [4] developed a methodology that integrates the use of multi-source satellite data along with existing geospatial data to assess the impact of floods that occurred in Sindh. In this research, the remote sensing data is processed with the help of Google Earth Engine. In addition, the spatial analysis is conducted with the help of ArcGIS. The floods that struck Pakistan in 2022 are the main subject of this study. Thousands of communities and 25,000 km^2 of land were impacted by the floods.The research concludes that the use of satellite data along with existing geospatial data may lead to the accurate and quick assessment of the impact of the floods.

Avioz et al. [5] examined the application of data fusion using Sentinel-2 satellite images and UAV photography in the prediction of canopy nitrogen content (CNC) in citrus crops. The researchers used various spectral and textural features, such as those derived using Structure from Motion Photogrammetry, from the images of the leaf samples and the UAV photographs. The researchers developed a Random Forest algorithm for the prediction of CNC. The researchers found that the performance of the model was higher than the individual models, with an R^2 of 0.80 and an RMSE of 0.17 kg/m^2. The research demonstrates the importance of using satellite and UAV images in the estimation of nitrogen content in citrus crops for sustainable management.

Fan et al. [6] developed a dataset on soil moisture for the whole world with a spatial resolution of 1 km. Using the simulation results for the backscatter signal from SAR, the authors proposed a method for assessing soil moisture by means of a dual polarization algorithm (DPA). The dataset has a frequency ranging from three to twelve days. The whole land surface area of the world is covered by the dataset from 2016 to 2022. The validation results indicated a good agreement with ubRMSD = 0.077 m^3/m^3.

Raza et al. [7] proposed a data fusion approach based on machine learning and remote sensing techniques for forecasting the yield of corn crops. The data used in their research included Sentinel-2 and Landsat 8 vegetation data, soil data and topographical data obtained through LiDAR technology. The best-performing machine learning algorithm, after testing several algorithms, was found to be the Random Forest algorithm, where the R2 value was found to be greater than 75%. The research showed that the accuracy of yield prediction in precision agriculture could be improved significantly.

Sabir et al. [8] used multisensor SAR imagery from EOS-04 and Sentinel-1 for deforestation monitoring in Sub-Himalayan forests. The research mapped deforestation in Haldwani forests using a statistical method of change detection, termed the "Cumulative Sum Control Chart." Both VH polarization of Sentinel-1 and HV polarization of EOS-04 performed in a similar manner, as revealed by the results, with an accuracy of approximately 0.71. While discussing the precise identification of changes in forests, the study highlights the need for polarization channels, resolution, and satellite availability.

Teixeira et al. [9] examined 64 different studies between 2016 and 2024 as part of a thorough review of the topic of a comprehensive review of soil moisture estimate using remote sensing for agricultural purposes. This included different types of sensors, satellites, and unmanned vehicles. According to the review, the most frequently used techniques for the prediction of soil moisture were the learning-based techniques using machine and deep learning. This review reaches a conclusion that more precise soil moisture measurement using remote sensing and other auxiliary data for precision agriculture.

Mkhwenkwana et al. [10] offered a critical analysis of the use of the Sentinel satellite data sets for the measurement of crop production soil moisture. The data sets of the different

satellites were used. According to the paper, the use of machine learning models like RF, SVR, CNN, etc., with the SAR polarization data sets and the multispectral index data sets like NDVI and NDMI is very common for the purpose of soil moisture estimate. However, the accuracy of the results may be affected by certain factors like terrain complexity and vegetation density. In order to improve the process of soil moisture estimate for the purpose of agriculture, the use of advanced machine learning models with the data sets of the satellites is of critical importance.

Li et al. [11] employed the technique of data fusion using the multi-source remote sensing data fusion based on the UAV for the estimation of the soil water content (SWC) in the cotton crop. The study employed the combination of meteorological, multispectral, and thermal infrared data from the imagery collected using the UAVs. After the evaluation of various machine learning algorithms, the best-performing algorithm for the estimation of the SWC was CatBoost, where the R2 value was 0.762. The characteristics of the estimation of the SWC in the cotton crop were based on the vegetation and drought indices.

Lu et al. [12] recommended that hydrological modeling and data fusion of remote sensing information could be used to simulate BW and GW in the Xiangjiang River Basin. To improve water resource assessment results, calibration systems were integrated with remote sensing evapotranspiration information. The accuracy of the integrated ET dataset was greater in comparison to other datasets used alone. The modeling of BW and GW simultaneously was significantly improved through the dual variable calibration method, which also reduced uncertainty in model results. This strategy will improve watershed management, water resource allocation, and ecosystem management.

Lin et al. [13] proposed the technique that combines hyperspectral photos and Synthetic Aperture Radar data. The technique used a Convolutional Neural Network to predict soil salinity based on 123 soil samples collected in Northeast China. The technique also used an MSD-WGAN-GP model to combine hyperspectral photos and SAR data. The proposed technique increased R2 values while reducing RMSE compared to hyperspectral photos alone. This research indicates that it is possible to increase the accuracy of estimating soil salinity by fusing data collected by various remote sensing techniques.

Rahmati et al. [14] studied the soil moisture retrieval methods using sentinel-1 data with over a decade of operation.The paper focuses on the advantages of using SAR technology to detect soil moisture regardless of the weather. The authors also mention the latest developments, like the use of AI technology to detect soil moisture, as well as SAR interferometric techniques. However, there are still challenges like surface roughness, vegetation, and models. To improve the use of multi-sensor technology to detect soil moisture, the authors suggest combining sentinel-1 with sentinel-2 and SMAP.

Irannejad et al. [15] developed an explainable deep learning framework for integrating climate and multi-source remote sensing data to estimate the AGB of the forest. The authors used meteorological and topography factors derived from LiDAR with the aid of Sentinel-1, Sentinel-2, and Landsat-8 data. In the prediction of the biomass, various deep learning models were compared, including the U-Net3+ model. The U-Net3+ model was the best. The results showed the importance of explainable AI models in the prediction of the biomass.

III. METHODOLOGY

A. Dataset Description

The soil moisture dataset used in the research was created based on multi-source satellite data provided by Google Earth Engine. The dataset was created by integrating data from various satellite sources such as Sentinel-1, Sentinel-2, and MODIS satellite data to estimate the soil moisture levels in various districts of the state of Tamil Nadu. The satellite data provide various parameters related to the environment and geographical data, which are strongly related to the soil moisture levels.

The collected dataset has various parameters such as EVI, Elevation, Land Surface Temperature, NDMI, NDVI, NDWI, Rainfall, SAVI, Slope, VH, VV, VV/VH ratio, latitude, longitude, and district, which are related to the soil moisture levels. The soil moisture level is the parameter used to train the model

The data is created based on the satellite data provided by Google Earth Engine, which is then exported to the system to analyze the data. 80% of the data is utilized to train the model, while the remaining 20% is used to test it.

Data preprocessing methods like feature scaling and data normalization are used to train the model more precisely. The model is then used to predict the soil moisture levels after the data has been standardized.

B. Feature-Level Data Fusion for Soil Moisture Prediction

While employing multi-source satellite data to enhance the accuracy of soil moisture prediction, the data fusion at the feature level is of prime importance. In the proposed research, a unified feature representation is achieved through the integration of various environmental attributes from the satellite data of Sentinel-1, Sentinel-2, and MODIS satellites. Complementary data such as radar backscatter, vegetation indices, rainfall, and land surface temperature, which are all strongly correlated with soil moisture levels, are provided through the satellite data.

Random Forest, XGBoost, and Deep Neural Network (DNN) are the various machine learning and deep learning algorithms employed to train the data. These algorithms are capable of accurately predicting the soil moisture levels through the detection of the intricate relationships between the environmental attributes and the soil moisture levels, thus enhancing the reliability of the soil moisture estimation through

the data fusion at the feature level.

1. Feature Fusion Representation:

The combined feature vector used for soil moisture prediction is represented as:

$$F = (EVI, Elevation, LST, NDMI, NDVI, NDWI, Rainfall, SAVI, Slope, VH, VV, \frac{W}{VH}, Latitude, Longitude) \quad \text{…….. (1)}$$

2. Soil Moisture Prediction Model:

The prediction model learns the relationship between the fused feature vector and soil moisture values.

$$SM = f(F) \quad \text{……. (2)}$$

where SM represents the predicted soil moisture value and F represents the fused satellite feature vector used as input to the model.

C. *Deep Neural Network for Soil Moisture Prediction*

Learning complex nonlinear correlations between many environmental factors is the most common application of DNNs. In this study, the fusion characteristics of the satellite data from the Sentinel-1, Sentinel-2, and MODIS satellites have been utilized for the prediction of soil moisture using the DNN model. The environmental factors that the DNN model accepts as inputs are EVI, NDVI, NDMI, NDWI, rainfall, elevation, slope, LST, VV, VH, and VV/VH Ratio.

The structure of the DNN model comprises the input layer, several hidden layers using the ReLU activation function, and the output layer for predicting the soil moisture levels. These hidden layers in the DNN model have learned the correlation between the soil moisture levels and the environmental factors through the extraction of significant patterns from the fused satellite data.

The predicted soil moisture value is computed as:

$$SM = f(WX + b) \quad \text{…….. (3)}$$

where SM represents the predicted soil moisture, X represents the input feature vector, W represents the weight matrix, and b represents the bias term of the neural network.

D. *Random Forest Model*

An ensemble technique referred to as the Random Forest method is a combination of decision trees. The technique is useful for increasing the accuracy of the prediction results. In the current study, the Random Forest model is applied to examine the relationship between environmental factors and the levels of soil moisture.

The prediction of the soil moisture is obtained by averaging the results of all the decision trees. This improves the accuracy of the results.

The prediction of the Random Forest method can be represented as:

$$SM = \frac{1}{N}\sum_{i=1}^{N} T_i(X) \quad \text{……. (4)}$$

where N represents the number of trees and $T_i(X)$ represents the prediction from the i-th decision tree.

E. XGBoost Model

Another sophisticated ensemble technique, XGBoost, or Extreme Gradient Boosting, involves the sequential building of decision trees to reduce errors in the model's predictions. It improves the model's performance through gradient boosting, which corrects the errors of the preceding trees.

XGBoost has been employed to establish the complicated relationships existing between satellite features and soil moisture readings. It has been trained to optimize a loss function to enhance the accuracy of the model's predictions.

The following is a representation of the XGBoost prediction model:

$$SM = \sum_{k=1}^{k} f_k(X) \quad \text{..... (5)}$$

where K represents the number of boosting trees and $f_k(X)$ represents the prediction function of each tree

Layer (type)	**Output Shape**	**Param #**	**Connected to**
input_layer_3 (Input Layer)	(None, 18)	0	-
dense_18 (Dense)	(None, 128)	2,432	input_layer_3[0]
batch_normalization (BatchNormalization)	(None, 128)	512	dense_18[0][0]
dense_19 (Dense)	(None, 128)	16,512	batch_normalizat…
batch_normalization (BatchNormalization)	(None, 128)	512	dense_19[0][0]
dense_20 (Dense)	(None, 128)	16,512	batch_normalizat…

Table.1 Model summary for Ensemble

IV. WORKFLOW

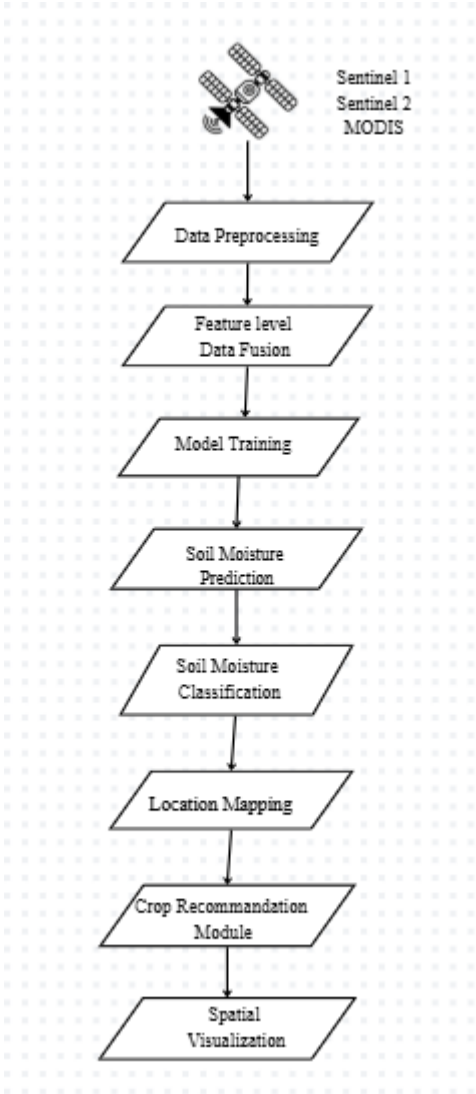

Figure.1 Workflow

Figure 1 illustrates the workflow of the proposed system for the prediction of soil moisture. The system begins with the collection of data from various sources, such as Sentinel-1, Sentinel-2, and MODIS satellites, along with environmental parameters like NDVI, NDMI, rainfall, land surface temperature, elevation, slope, radar backscatter, and geographic location. The collected data then goes through the preprocessing step, where the data is cleaned, filled, and normalized. The data is then passed to the next step, where the data fusion technique is applied to merge the features of the satellite images with the environmental parameters. The combined data is used as the input to the training of the models to estimate the soil moisture. We use three models: Random Forest, XGBoost, and Deep Neural Network. The results are categorized into four types of soil moisture: Dry, Low, Medium, and High. The location of the crops is identified by the geographic location of the data. The location is then mapped to provide the crops to be grown on the basis of the identified location. The results are then visualized using the spatial visualization technique. The maps are created using the geospatial mapping tools.

V. RESULT AND DISCUSSION

Predictions for soil moisture using variables and the models used were evaluated using regression metrics developed for this purpose: RMSE, MAE, and R2. Random Forest outperformed each of the other models evaluated based upon the metrics-RMSE=0.00456, MAE=0.00249 and R2=0.9927. The results indicated a very strong correlation between environmental variables and soil moisture content when using the Random Forest model for prediction. XGBoost followed second-evaluated based upon RMSE=0.00462, MAE=0.00269, and R2=0.9925. The results provided sufficient evidence of a very good correlation between environmental variables and soil moisture content for the XGBoost model. The third model evaluated was the Deep Neural Network with an RMSE=0.00959, MAE=0.00640 and R2=0.9680 indicating it had moderate reliability to predict soil moisture content based on environmental variables. Finally, when the three independent models were combined-Random Forest, XGBoost, and Deep Neural Network-the resulting model provided RMSE=0.00525, MAE=0.00339, and R2=0.9904 as compared to independent the three models

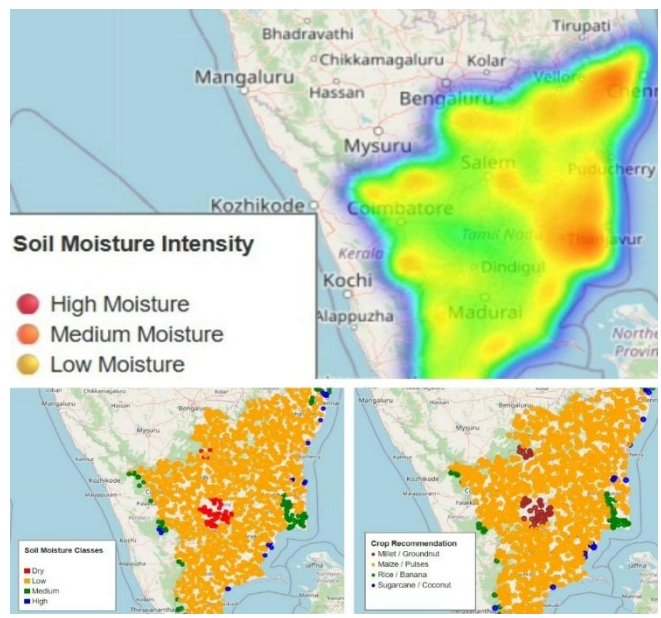

Figure.2 Spatial Soil Moisture Intensity Map Across Tamil Nadu

Figure 2 depicts the expected soil moisture values as related to a heatmap of the soil moisture intensity for the state of Tamil Nadu, using a color scale to show regions of high soil moisture represented in red, medium soil moisture

represented in orange, and low soil moisture represented in yellow, thereby showing the distribution of the various soil moisture conditions across the state of Tamil Nadu.

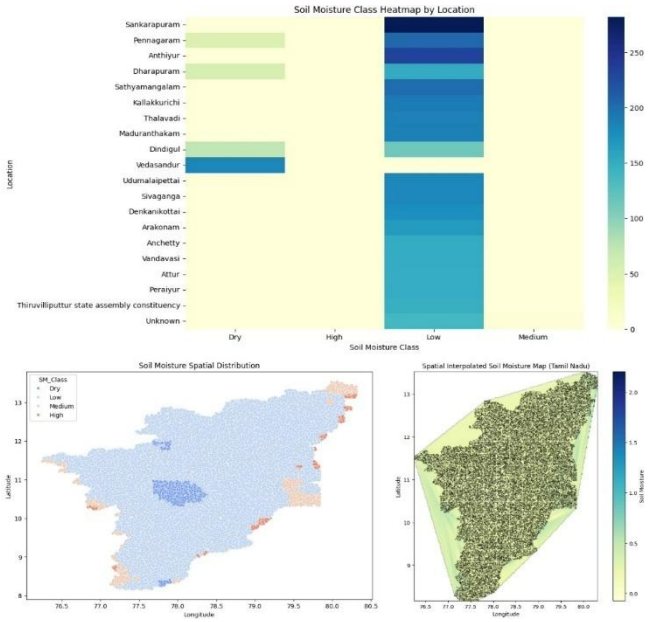

Figure.3 Soil Moisture Classification and Crop Recommendation Map

As we can see in Figure 3, there is a collage-like presentation of spatial maps. The left side illustrates how the soil moisture is distributed across Tamil Nadu. The levels of soil moisture are classified into dry, low, medium, or high.

The right side is a presentation of possible crop selection based on the expected levels of soil moisture. The various shades represent different types of crops that can be planted, including millet, groundnuts, maize, legumes, rice, bananas, sugarcane, and coconut.

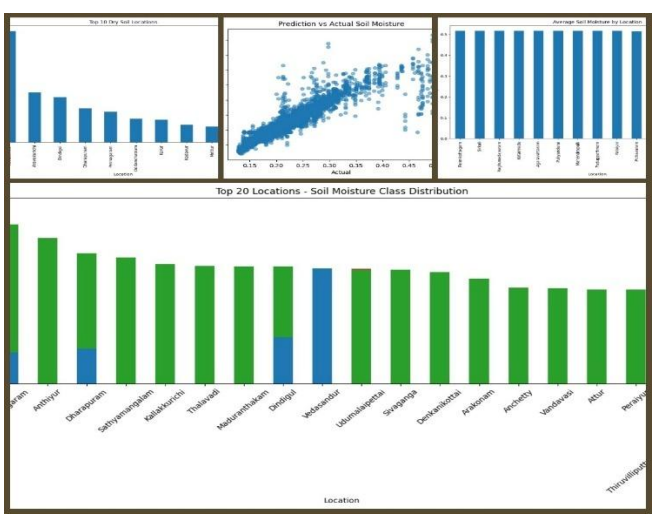

Figure.4 Location-Based Soil Moisture Analysis and Spatial Interpolation

The set of figures says three types of analytical visual representations to describe how soil moisture is distributed over the area. In the figure on the layout (Figure 4), the top heat map allows you to compare levels of soil moisture across various areas.

The map on the lower left part of the figure depicts categories of soil moisture levels, which are dry, low, medium, and high, using different shades to describe how these categories are distributed geographically over Tamil Nadu.

The map on the bottom right part of the figure depicts a geospatial interpolation of soil moisture levels grouped into various clusters, which gives an interpolated surface of how soil moisture is distributed over various agro-ecological areas of Tamil Nadu.

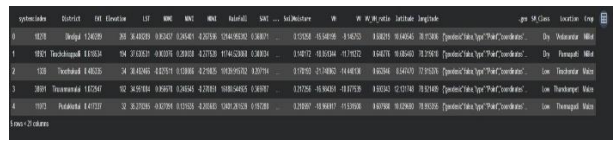

Figure 5 Statistical Analysis of Soil Moisture Predictions Across Locations

Figure 5 summarizes various analytical charts into one composite chart that can be used to estimate results for soil moisture content. The first chart depicts the 10 driest soil locations suggesting greater opportunity for good irrigation management. The second chart shows the predicted amount of soil moisture with actual amounts where the correlation is nearly linear and the points are closely clustered around a diagonal line of corresponding points indicating a strong relationship between the predicted and measured amounts. The third chart represents average soil moisture content at different locations to easily see which locations have high or low amounts of soil moisture. The final chart represents the soil moisture classes for the 20 highest soil moisture locations indicating the degree of variability in the amount of soil moisture at different locations.

VI. CONCLUSION AND FUTURE WORK

The proposed work presents a feature-level data fusion technique that is used to forecast soil moisture levels through the incorporation of various satellite and environmental data sets. It uses various features such as NDVI, NDMI, rain, land surface temperature, elevation, slope, and radar backscatter (VV and VH). It also employs various machine learning algorithms such as a Random Forest, XGBoost, and a Deep Neural Network. These algorithms are then fused using ensemble technique to improve its accuracy. It is able to achieve RMSE of 0.00526, MAE of 0.00339, and R^2 of 0.9904. It also creates soil moisture maps to help farmers in Tamil Nadu. Future work is planned to incorporate more satellite data sets and also test other advanced deep learning algorithms to improve its accuracy. It is also possible to evolve it to a web or mobile-based decision support tool that can be used to help farmers.

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